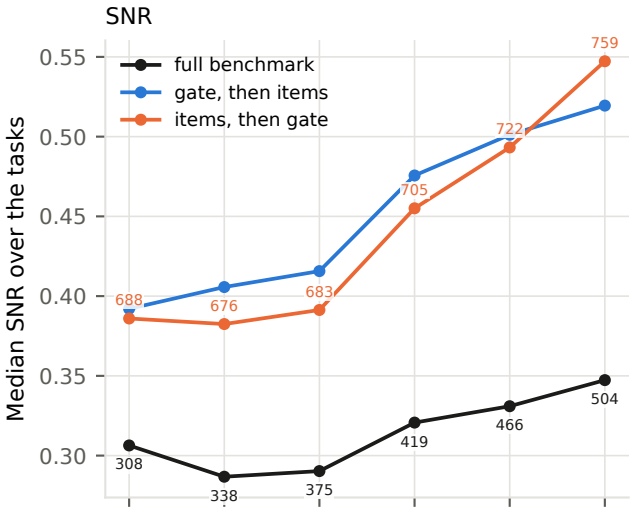
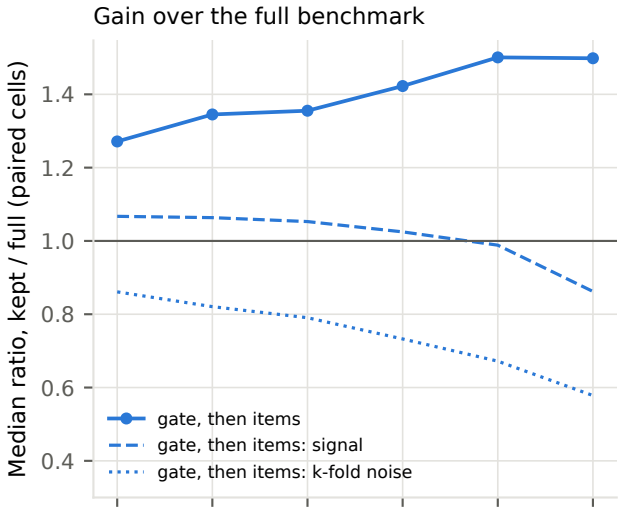
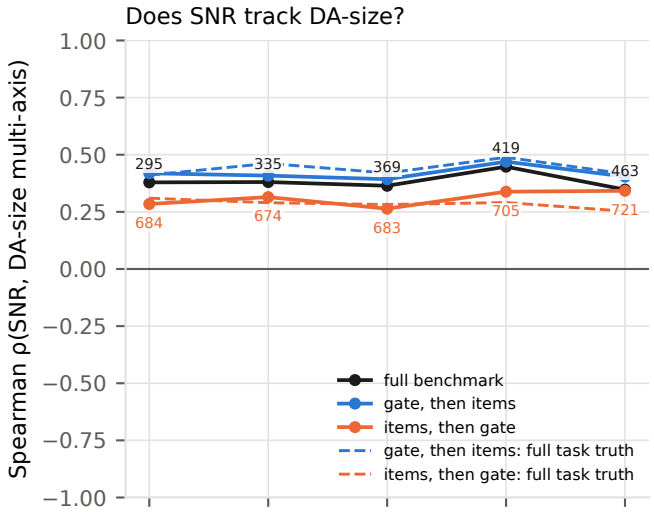

SNR
full benchmark
gate, then items
items, then gate
Median SNR over the tasks
0.55
0.50
0.45
0.40
0.35
0.30
688
676
683
705
722
759
308
338
375
419
466
504
90M
175M
350M
600M
1B
1.7B
Model size


Gain over the full benchmark
Median ratio, kept / full (paired cells)
1.4
1.2
1.0
0.8
0.6
0.4
gate, then items
gate, then items: signal
gate, then items: k-fold noise
90M
175M
350M
600M
1B
1.7B
Model size


Does SNR track DA-size?
Spearman ρ(SNR, DA-size multi-axis)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
295
335
369
419
463
684
674
683
705
721
full benchmark
gate, then items
items, then gate
gate, then items: full task truth
items, then gate: full task truth
90M
175M
350M
600M
1B
Model size